# CST563 - Advanced Machine Learning

## CST 563 (Advanced Machine Learning) Syllabus

### Course Information

#### Course Details

- **Course Title:** Advanced Machine Learning
- **Course Number:** CST 563
- **Meeting Location:** Physical
- **Units:** 4

#### Course Description

In this course students learn to use advanced machine learning methods. The advanced machine learning methods include support vector machines, dimensionality reduction, ensemble methods, methods for machine learning with time series data, and especially neural networks.

#### Materials

- Course textbook: "Pattern Recognition and Machine Learning" by Christopher M. Bishop

#### Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system - GPU acceleration for machine learning computations (available through Google Colab, cloud services like AWS/Azure/GCP, or local GPU hardware) - Machine learning frameworks and libraries (TensorFlow, PyTorch, scikit-learn) - Access to large dataset storage and processing capabilities

#### Course Objectives

At the end of this class, you should be able to:

1. Implement and apply support vector machines and dimensionality reduction techniques
2. Design and utilize ensemble methods for complex machine learning problems
3. Develop neural network architectures and apply deep learning techniques
4. Analyze and process time series data using advanced machine learning approaches

### Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

| Assignment Kind | Value |
|---|---|
| Projects | 35% |
| Exams | 35% |

| Assignment Kind | Value |
|---|---|
| Participation | 30% |

**Projects**

Projects will take the form of neural network implementations, comparative analysis of ML algorithms on large datasets, ensemble method development, time series prediction models, and independent research on advanced ML techniques. Students will work with real-world datasets and deploy models in production-like environments.

**Exams**

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

**Participation**

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

**Grade Assignment**

Grades as assigned based on the standard ranges, which are outlined below.

| Grade Range | Letter Grade |
|---|---|
| $[97, \infty)$ | A+ |
| $[93, 97)$ | A |
| $[90, 93)$ | A- |
| $[87, 90)$ | B+ |
| $[83, 87)$ | B |
| $[80, 83)$ | B- |
| $[77, 80)$ | C+ |
| $[73, 77)$ | C |
| $[70, 73)$ | C- |
| $[60, 70)$ | D |
| $[0, 60)$ | F |

## Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.